

QUALITY RISK ASSESSMENT & PRODUCT IMPACT EVALUATION

Addendum to Nonconformance Report NCR# N26436

NCR Number: N26436	Date of Evaluation: 26-Aug-2026 / 27-Aug-2026	Department: Microbiology (Sterility)
Affected Location: Cleanroom Suite 114 (Anteroom 114A & Buffer Room 114B)	Missed Period: 02-Aug-2026 to 15-Aug-2026 (2 Weeks)	Applicable SOPs: MICRO-SOP-2, CORP-SOP-4, SOP 2.100.019

1. Executive Summary & Nonconformance Scope

During the supervisory review of environmental monitoring records, it was identified that routine weekly surface sampling (RODAC contact plates) and active air sampling for **Cleanroom Suite 114** (Anteroom 114A ISO 8 and Buffer Room 114B ISO 7) were inadvertently omitted by EM analyst SMO for two consecutive weekly monitoring cycles covering **02-Aug-2026 through 15-Aug-2026**, contrary to the requirements of *MICRO-SOP-2 (Environmental Monitoring of the Clean Room Facility)*. The primary root cause was determined to be analyst oversight during high-throughput testing rotations, compounded by the absence of a real-time end-of-week EM schedule reconciliation tracking mechanism.

Because three (3) Celsis sterility testing Out-Of-Specification (OOS) investigations were initiated for sample submissions processed in **Cleanroom 114B** during this exact timeframe, this formal Quality Risk Assessment was conducted to evaluate the state of cleanroom control, and establish defensible mitigation rationale for each correlated OOS investigation.

2. Correlated Sterility Out-Of-Specification (OOS) Investigations

A thorough reconciliation of all testing logs and room assignments in Cleanroom Suite 114 between 02-Aug-2026 and 15-Aug-2026 confirmed three (3) correlated sterility test failure investigations in **Room 114B**:

Submission ID (ETX)	OOS Number	Testing Room	Client / Sample Description	Test Date	Investigation Status
ETX-260729-0660	OOS-261824	Room 114B	Boudreaux's New Drug Store (E01607) - Methylcobalamin / Glycine	03-Aug-2026	Phase I in progress (Micro ID pending ECD 20-Aug-26)
ETX-260803-0212	OOS-261861	Room 114B	Grand Avenue Pharmacy 13409 (E13409) - Triamcinolone Acetonide	03-Aug-2026	Phase I in progress (Micro ID ETX-260813-0684 ECD 24-Aug-26)
ETX-260806-0456	OOS-261877	Room 114B	Celsis Sterility Batch (Hold as of 17Aug26)	06-Aug-2026	Phase I Lab Investigation in progress

Note on Excluded Records: Formal room allocation verification confirmed that **OOS-261878** and **OOS-261932** were processed inside Cleanroom Suite 115 (not Suite 114) and are therefore completely unaffected by this nonconformance and excluded from this evaluation.

3. Scientific Risk & Barrier Analysis (State of Environmental Control)

To evaluate whether the missed weekly background EM in Cleanroom Suite 114 contributed to or caused these three (3) sterility test failures, a multi-barrier quality evaluation was conducted:

3.1 Lack of Widespread Contamination Trend

- **No Widespread Trend:** The emergence of only three (3) isolated failures confirms that no widespread, airborne, or systemic bioburden contamination existed in Cleanroom Suite 114 or Room 114B.

3.2 Primary Engineering Control (PEC / ISO 5 BSC) & Session In-Process Monitoring

- **Direct Critical Processing Zone Isolation:** All sample handling, vial manipulations, membrane filtration, and media inoculations were conducted exclusively within certified ISO Class 5 Biological Safety Cabinets (BSCs) located inside Room 114B. The ISO 5 unidirectional vertical laminar airflow provides continuous HEPA barrier protection against background room air.
- **Session-Specific In-Process EM Execution:** Routine session-specific environmental monitoring WAS fully executed for every individual testing process per MICRO-SOP-2 Section 8.1.3:
 - *Passive Air (Settling Plates):* 90 mm TSA plates exposed in the BSC during testing exhibited zero microbial growth (0 CFU/plate, Action Limit ≥ 1 CFU).
 - *Surface Contact Plates (RODAC):* Work surface contact plates sampled post-testing demonstrated no microbial recovery.
 - *Personnel Monitoring:* Left and right gloved fingertip touch plates (LH/RH) for processing analysts met all alert/action limits.

3.3 Continuous Facility Automated Engineering Controls (Lighthouse Monitoring)

- **24/7 Automated Sensor Surveillance:** Physical environmental parameters for Cleanroom Suite 114 (Anteroom 114A and Buffer Room 114B) are continuously monitored and logged electronically by the automated Lighthouse System per SOP 2.700.014.
- **Zero Physical Excursions:** Retrospective audit of Lighthouse continuous data from 02-Aug-2026 through 15-Aug-2026 confirmed continuous positive pressure cascades (ISO 5 BSC -> ISO 7 Buffer Room 114B -> ISO 8 Anteroom 114A -> General Facility) and tight temperature/humidity control (66°F–72°F / 30%–60% RH) with no HVAC shutdowns or pressure reversals.

3.4 Cleaning, Disinfection, and Material Transfer Regimen

- Daily and weekly cleaning and sporicidal disinfection protocols (acidified bleach solution and sterile 70% IPA per SOP 2.600.018) were executed and documented according to schedule.
- Staged sanitization (10-minute sporicidal contact time) for all sample bins, consumables, and media bottles prior to cleanroom and BSC introduction remained fully in effect, preventing ingress of external bioburden.

3.5 Method Negative Controls & Reagent Verification

- All method negative controls (media blanks, diluent soak blanks, and system rinse controls) processed concurrently with the affected submissions demonstrated valid negative results (baseline RLU on Celsis luminometer), confirming reagent sterility and lack of systemic assay contamination.

3.6 Data Gap Mitigation via Microbial Identification & Post-Event Reinstatement

- **Microbial Identification (Micro ID):** For the three OOS cases in Room 114B (OOS-261824, OOS-261861, OOS-261877), recovered microbial isolates are being identified to genus/species to cross-reference against cleanroom baseline flora. Isolates distinct from cleanroom or skin commensals provide definitive evidence of client product container origin.
- **Reinstatement Sampling:** Routine weekly surface and active air EM for Suite 114 was immediately resumed starting 17-Aug-2026, meeting all alert and action limits (Anteroom <50 CFU, Buffer Room <5 CFU), confirming no persistent bioburden accumulation occurred.

4. Impact Assessment & Risk Scoring Matrix (Closure by Quality)

Risk Parameter	Assigned Score	Grading Criteria & Rationale
Frequency	2	Between 2 to 10 historical occurrences (isolated supervisory finding across monitoring cycles).
Severity	2	Changes in processes/systems with no retroactive implications; affected batches quarantined under OOS.
Magnitude	2	>2 to <10 Process/Material/Equipment Affected (3 correlated sterility OOS investigations in Room 114B under review).
Overall Grading Score	8	Score = Frequency (2) × Severity (2) × Magnitude (2) = 8
Impact Level	MAJOR	Major Impact Category (Score Range 3 to 17) — appropriately categorized.

5. Corrective & Preventive Action (CAPA) Summary

1. **Immediate Containment & Reinstatement:** Routine weekly surface and active air EM for Suite 114 was fully reinstated with passing results.
2. **Tracking Sheet Implementation:** Microbiology Supervisor Robin Seymour instituted a mandatory Weekly EM Sampling Tracking Sheet (*EM Monitoring of Cleanrooms Rotation Schedule.xlsx*) requiring weekly reconciliation and supervisory verification.
3. **Personnel Retraining:** Analyst SMO completed documented Read and Understood retraining on MICRO-SOP-2 Section 8.2.
4. **CAR Requirement:** Based on effective root cause remediation and intact environmental barriers, a formal Corrective Action Response (CAR) is **NOT** required.

6. Quality Unit Conclusion & Final Disposition

Based on the totality of evidence—including uninterrupted 24/7 Lighthouse physical parameter stability, satisfactory session-specific ISO 5 BSC settling and contact plate monitoring, routine cleaning and disinfection activities, acceptable analyst glove touch plates, valid method negative controls, and clean post-reinstatement EM results—the microbiological integrity and physical containment of Cleanroom Suite 114 (114A/114B) remained in a validated state of control during 02-Aug-2026 through 15-Aug-2026. The missed weekly sampling represents an isolated documentation and scheduling oversight without evidence of environmental breakdown. The three (3) correlated sterility OOS investigations in Room 114B remain valid, and each report shall cross-reference NCR# N26436 and incorporate this risk evaluation to support Phase I closure.

Evaluated / Prepared By:	Reviewed By:	Quality Assurance Approval:
<u>Oiyue Chen, Ph.D.</u> Project Microbiologist Date: 27-Aug-2026	<u>Robin Seymour</u> Microbiology Laboratory Supervisor Date: _____	<u>Quality Unit Designee</u> Quality Assurance Date: _____